

Programming Example: List connected VISA compatible resources using PyVISA

August 30, 2018

PyVISA is a software library that enables Python applications to communicate with resources (typically instruments) connected to a controlling computer using different buses, including: GPIB, RS-232, LAN, and USB.

This example scans and lists the available resources.

It requires PyVISA to be installed (see the PyVISA documentation for more information)

```
#Example that scans a computer for connected instruments that
#are compatible with the VISA communication protocol.
#
#The instrument VISA resource ID for each compatible instrument
#is then listed.
#
#
#Dependencies:
#Python 3.4 32 bit
#PyVisa 1.7
#
#Rev 1: 08302018 JC
```

```
import visa

def main():
    rm = visa.ResourceManager()
    print (rm.list_resources())

if __name__=='__main__':
    main()
```

Here is the code:



And here is the result of a scan:



Each connected instrument returns a specific formatted string of characters called the VISA Resource ID.

The resource ID format is as follows:

'Communication/Board Type (USB, GPIB, etc.)::Resource Information (Vendor ID, Product ID, Serial Number, IP address, etc.)::Resource Type'

In the response, each resource is separated by a comma. So, we have three resources listed in this example:

'USB0::0x0483::0x7540::SPD3XGB4150080::INSTR' – This is a power supply (SPD3X) connected via USB (USB0)

'USB0::0xF4EC::0x1301::SVA1XEAX2R0073::INSTR' – This is a vector network analyzer (SVA1X) connected via USB (USB0)

'TCPIP0::192.168.55.122::inst0::INSTR' – This is an instrument connected via LAN using a TCPIP connection at IP address 192.168.55.122



North American Headquarters

SIGLENT Technologies NA
6557 Cochran Rd Solon, Ohio 44139
Tel: 440-398-5800
Toll Free: 877-515-5551
Fax: 440-399-1211
info@siglent.com
www.siglentamerica.com/

European Sales Offices

SIGLENT TECHNOLOGIES GERMANY GmbH
Staetzlinger Str. 70
86165 Augsburg, Germany
Tel: +49(0)-821-666 0 111 0
Fax: +49(0)-821-666 0 111 22
info-eu@siglent.com
www.siglenteu.com

Asian Headquarters

SIGLENT TECHNOLOGIES CO., LTD.
Blog No.4 & No.5, Antongda Industrial Zone,
3rd Liuxian Road, Bao'an District,
Shenzhen, 518101, China.
Tel: + 86 755 3661 5186
Fax: + 86 755 3359 1582
sales@siglent.com
www.siglent.com/ens